

U.S. Multi Industry & Electrical Equipment

Liquid Cooling Primer: Cold Plates



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As data center rack thermal densities continue to rise driven by the rapid scaling of AI workloads and increasing power density of GPUs, air cooling alone is no longer sufficient to manage the heat generated by modern compute architectures. As a result, direct-to-chip (DTC) liquid cooling has become a critical part of the data center ecosystem, with **cold plates emerging as a key enabler** of the transition (from air cooling to liquid cooling). At the core of the liquid cooling ecosystem, two components drive the thermal loop - CDUs (see our [earlier primer](#)) and cold plates. Cold plates perform the most important task in the architecture: **physically transferring the heat from the chip surface to the coolant**, and are the primary mechanism enabler in the liquid cooling stack.

The importance of cold plates becomes apparent when viewed in the context of modern AI data center racks. For example, a single NVIDIA GB200 NVL72 **rack can contain over 100 individual cold plates** (one for each GPU / CPU), highlighting the scale at which they are deployed. As rack densities continue to increase with next-gen GPUs expected to reach multi-thousand kW TDP levels, the ability of cold plates to efficiently manage heat flux and thermal resistance becomes increasingly critical to overall system performance. Despite this integral role, we view cold plates through a more nuanced lens than other liquid cooling components; structural tailwinds supporting volumes are offset by commoditization and possibly obsolescence.

On one hand, the near-term outlook is clearly favorable. Adoption of liquid cooling is accelerating, hyperscalers are committing to fully liquid-cooled architectures (NVIDIA announcement for Rubin), and cold plates are effectively a **non-optional component** within DTC systems. This creates strong volume tailwinds, supported by increasing liquid cooling penetration (from ~30–40% of new additions today towards majority adoption by 2030). On the other hand cold plates (unlike CDUs) are inherently simpler components, with value largely concentrated in design rather than system-level complexity. Service attach is comparatively lower vs. CDUs, and cold plate OEMs largely capture hardware margins which are harder to defend than service margins. This raises the question of **how defensible the business model is over time**, particularly as designs commoditize, manufacturing scales, and the innovation / scarcity premium erode. And in extreme scenarios, technology such as direct-to-die have the ability to eliminate cold plates entirely.

We size the base case cold plate market to be ~\$6 - 7B by 2030 (growing from \$2 - 3B today). As with any data center equipment, this is highly dependent on your assumptions around GW added. Reach out to the authors or your Bernstein sales counterpart for the raw model in case you'd like to adjust assumptions.

In this note, we lay out a comprehensive primer on cold plates, including **how they work, what drives performance, key design trade-offs, and where innovation is occurring today**. We also assess the evolving competitive landscape, examine the economic model (including limited services revenue opportunity), and present a bottom-up view on market sizing and growth. Finally, we outline our perspective on the trajectory of the space, where cold plates remain a critical enabler of AI infrastructure growth in the near-term, but face increasing commoditization and disruption risks over a longer time horizon.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	2 Jul 2026	Price	TTM	Adjusted EPS				Adjusted P/E (x)		
			Closing	Target	Rel. Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
NVT (nVent)	O	USD	152.15	218.00	88.0%	USD	3.34	4.79	6.19	45.5	31.8	24.6
VRT (Vertiv)	O	USD	300.53	416.00	122.5%	USD	4.20	6.52	9.21	71.6	46.1	32.6
JCI (Johnson Controls)	O	USD	140.76	176.00	14.8%	USD	3.78	5.06	6.04	37.2	27.8	23.3
TT (Trane)	O	USD	478.13	550.00	(8.7)%	USD	13.06	14.96	17.61	36.6	32.0	27.1
CARR (Carrier)	M	USD	70.07	75.00	(26.1)%	USD	2.57	2.81	3.20	27.2	24.9	21.9
ETN (Eaton)	O	USD	398.52	534.00	(7.9)%	USD	12.07	13.29	16.32	33.0	30.0	24.4
SU.FP (Schneider)	O	EUR	275.90	310.00	2.2%	EUR	8.43	10.22	11.95	32.7	27.0	23.1
SPX			7,483.24									
EDME			1,622.79									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

We rate **VRT** Outperform with a target price of \$416.

We rate **NVT** Outperform with a target price of \$218.

We rate **CARR** Market-Perform with a target price of \$75.

We rate **TT** Outperform with a target price of \$550.

We rate **JCI** Outperform with a target price of \$176.

We rate **ETN** Outperform with a target price of \$534.

We rate **SU.FP** Outperform with a target price of €310.

DETAILS

EXHIBIT 1: Cold plate top-down market sizing

Year	2026E	2030E (Bear)	2030E (Base)	2030E (Bull)
GW added in year ¹	~20 GW ×	~40 GW ×	~40 GW ×	~40 GW ×
% share of liq. cooling	~30 - 40 % ×	~50 - 60 % ×	~60 - 70 % ×	~70 - 80 % ×
# of GPUs / GW ²	~1 - 2 M ×	~0.3 - 0.5 M ×	~0.3 - 0.5 M ×	~0.3 - 0.5 M ×
# of cold plates / GPU	1 ×	1 ×	1 ×	1 ×
Multiplier ³	1.2 ×	1.2 ×	1.2 ×	1.2 ×
ASP per unit ⁴	~\$150 - 300 ▼	~\$150 - 300 ▼	~\$400 - 600 ▼	~\$500 - 750 ▼
Cold Plates Market Size	\$2 - 3B	\$2 - 3B (flat CAGR)	\$6 - 7B (~20-30% CAGR)	\$8 - 10B (~35-40% CAGR)

1. Sourced from McKinsey 2. Assuming a range of 500-1000kW per GPU (split among Blackwell/Hopper) for 2026, range of 2000-3000kW per GPU (basis Rubin estimated) for 2030 3. To account for replacement / repair-based demand to materialize by end of decade 4. Estimated basis 0.30 \$/kW cooled (2026), flat prices (2030, bear case), 0.20\$/kW (2030, base case), 0.25\$/kW (2030, bull case); we have built a conservative estimate of a decline in \$/kW to bake in potential commoditization of cold plates

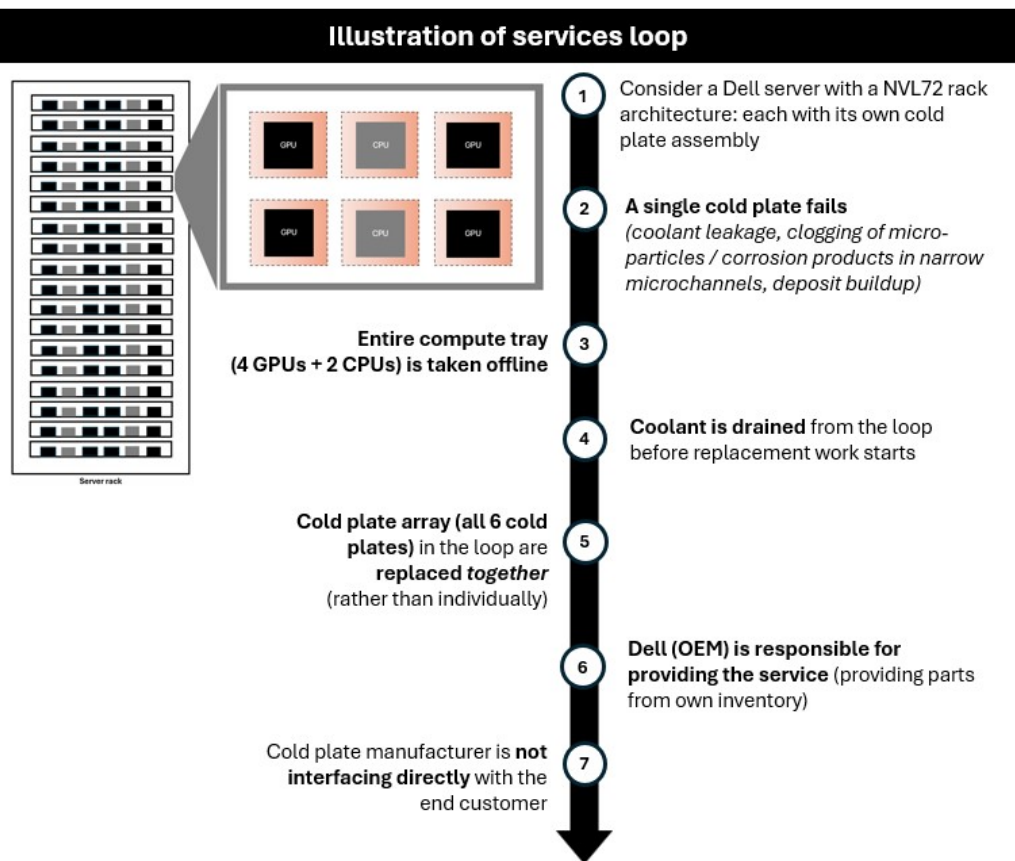
Source: Bernstein Analysis and Estimates

EXHIBIT 2: Overview of key cold plate players (non-exhaustive)

Company							
Year founded	1928	2001	2014	1992	1988	2022	2016
Country							 
Funding / acquisition price	\$9.5 billion (acquired, ~5.6x 2026 revenue)	\$4.75 billion (acquired)	NA	Self-funded	\$850 million (acquired, mid-single digit 2025 revenue Mx)	\$89 million (2 rounds)	\$111 million (3 rounds)
Key investors / acquirers	Eaton	Ecolab	Vertiv	Private	Schneider Electric	Johnson Controls, Legrand	Carrier, Mitsubishi, Samsung
Cold plates % of revenue	~40-50%	~40%	Mix b/w cold plates & heat exchangers	Small, but growing	Not available	Split b/w cold plates, CDUs, manifolds	Split b/w cold plates & CDUs
Commentary	Acquired by Eaton in Mar 2026 from GSAM Largest player globally having delivered 5 million+ cold plates to hyperscalers as of Sep '25	Acquired by Ecolab in Mar 2026 from KKR (15x return in 3y) Ecolab rationale to become a 'one-stop shop cooling solutions provider' to hyperscalers	Acquired by Vertiv in Apr 2026 Focus on high chip-density thermal problems , filling the gap between chip and facility-level cooling	33+ years of thermal manufacturing expertise based out of Taiwan Primarily sell to server OEMs instead of directly to hyperscalers	Schneider acquired 75% stake for \$850M Patented 'wide groove' tech with demonstrated capacity of up to 1000 W per GPU (distribute fluid top-down <i>simultaneously</i> across the chip)	Focused exclusively on two-phase cold plates with capacity of up to 4500 W per GPU Exclusively targeting next-gen NVIDIA Kyber rack architecture (vertically oriented server racks) Backed startups focused exclusively on two-phased DTC	Waterless, two-phased DTC cooling system with capacity of up to 600 W / GPU 'No pump' approach (unlike Accelsius) where phase change drives circulation by gravity differential

Source: Bernstein Analysis, Company Reports

EXHIBIT 3: Cold plate service attach relatively low; expect to see hardware margins

**Why is the services opportunity low?**

- 1 Cold plates designed for a **3–5-year lifecycle** (like servers) with low inherent failure rates, making service calls rare
- 2 Even when service is needed, it **flows through the OEM** (and not the cold plate manufacturer) under their service contracts with the customer
- 3 **OEMs consider cold plates to be commodity-level components** – customizations happen during server design stage, but servicing is standardized
- 4 **OEMs maintain inventory levels of server parts** and use the same to service any requests
- 5 If cold plate failures end up being very high, clauses in contracts push the manufacturer to get involved (but at a punitive level - not revenue generating)
- 6 As a result, most **cold plate manufacturers** continue to get **hardware margins** (and not service margins); margins lesser than CDUs due to lower component specialization

← All six cold plates are replaced together rather than individually, since opening the loop contaminates the TIM (thermal interface material) on all plates, and makes full replacement more cost-effective than incurring repeated labor for future failures →

Source: Bernstein Analysis, Expert Interviews

EXHIBIT 4: Cold plates attractive in near-term, but see real commoditization and obsolescence risk

Cold plates | Money to be made near-term, but high commoditization + tail obsolescence risk

Timeframe	Near-term (2026 – 2028)	Mid-term (2028 – 2030)	Long-term (2030+)
Risk of Commoditization	Low (Space is still evolving)	Moderate - High	High
Risk of Obsolescence	Low	Low	Moderate (dependent on etched silicon)
Commentary	We expect cold plates (in particular, single-phase bulk direct-to-chip) to be the main liquid cooling config. for the immediate future Expect 2-phase DTC to be commercialized this window increasing cooling capability Relatively mature and well-understood tech.; NVIDIA's reference design currently builds off this framework Some commoditization risk exists as specs get streamlined, but we think this is offset by the pace of change in the space	We think by 2030, 2-phase DTC will be the dominant cooling configuration We do believe there is high commoditization risk by 2030; two-phase tech. would have had time to mature and cold plates were commoditized components before the data center boom We do not think immersion cooling will take off as a serious alternative; it's too messy, requires large tanks that are tough to work on, and much more 2-phase refrigerants will be needed (which have GHG risk) Immersion also seems to have been de-prioritized by most hyperscaler roadmaps	If rack densities continue to increase, we think cold plates have a moderate risk of being a terminal business in the long-run Driven by possibility of direct silicon-etched cooling becoming a commercially viable config. while currently nascent there has been activity in the space (e.g., Microsoft-Corintis) and it removes cold plates while providing better cooling flux

We think cold plates have near-term tailwinds – but **there's a commoditization risk in the near-term (much higher than for CDUs)**

We expect 2-phase DTC would be relatively mature tech. by the end of the decade with well-established supply chains **eliminating the scarcity and innovation premium**

Cold plates have historically been consumables that are swapped out and do not need service; **many providers today (e.g., Accelsius) are primarily design shops with mfg. contracted to third parties**

As hyperscalers continue to spec. their products, we expect cold plates to be easy targets (given they will be needed in high volumes); **we expect to see a contract mfg. ecosystem drive this market**

Investments by large players (MSFT, TSMC) in silicon-etched microchannel cooling **can also eliminate the need for cold plates**; this is not a certain outcome but remains a risk to consider (especially as rack densities continue to increase)

Source: Bernstein Analysis

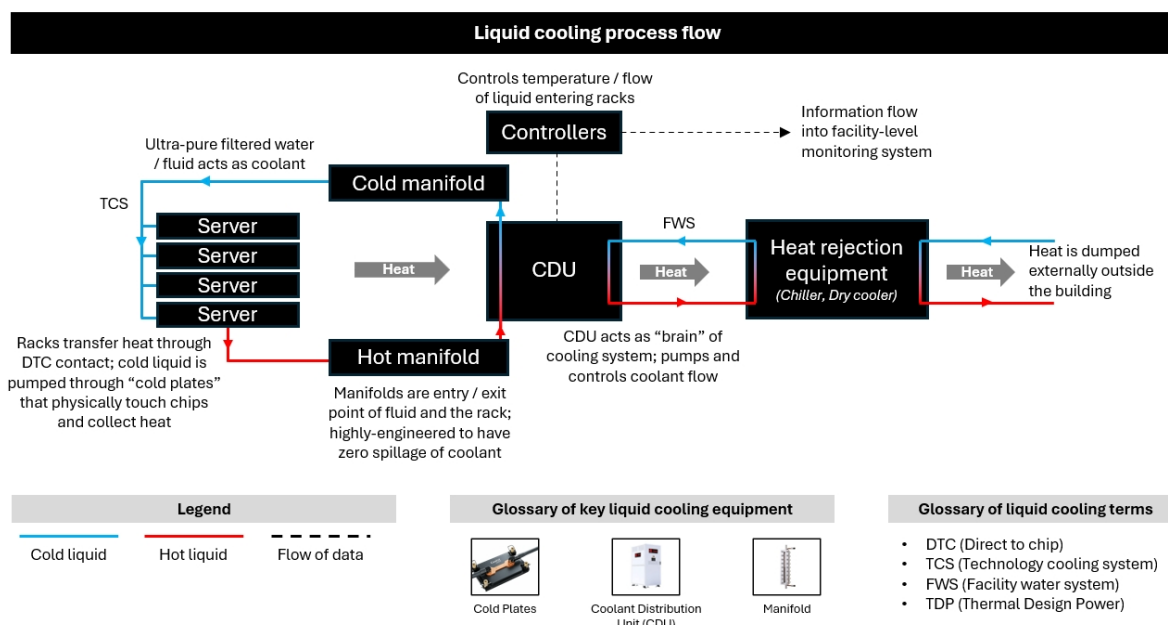
INTRODUCTION

In the past, a continuous flow of cool air over servers was sufficient to cool data centers. With time, chips and racks have grown increasingly dense, to the point where even freezing cold air blowing at gale force speeds wouldn't do the job. Needless to say, finding an alternative cooling modality became critical. And so, liquid cooling technology, which long sat at the fringe of data center infrastructure, suddenly found itself thrust into the spotlight. Looking ahead, rack densities and chip TDPs seem set to keep growing, forcing liquid cooling solutions to evolve with more rigorous server demands.

The predominant format of liquid cooling today is single-phase DTC (DTC stands for Direct-To-Chip). It involves cold plates (which are aptly named slabs of metal that have reduced temperature due to refrigerant that circulates *inside* them) coming into contact with GPU / CPU chips to extract heat. Coolant Distribution Units (CDUs) pump low temperature coolant (usually a water - propylene glycol mix) *through* the plates and recollect the spent liquid once they have extracted heat from the server. The heat is then extracted from the coolant (making it cold again) and the process repeats. It is called single-phase because the coolant stays in a single-phase (liquid) and does not change phases from liquid to gas via evaporation. This loop (i.e., circulating from the CDU through cold plates in the server) is called the TCS or Technology Cooling System. Once spent coolant reaches the CDU, it passes through a heat exchanger where it transfers heat to another cooling loop called the FWS or Facility Water System. The FWS connects to a chiller or a dry cooler to reject this heat outside a data center. However, as rack power densities continue to increase, single-phase DTC is seemingly reaching the theoretical limit of how much heat it can extract from the server rack.

Cold plates and CDUs are the most important to discuss. CDUs (Coolant Distribution Units) are essentially responsible for controlling the distribution, flow, and pressure of coolant across multiple rows and racks. Cold plates receive coolant from CDUs (via a manifold) and absorb heat through their surface before returning the spent coolant to the CDU. Both these components are seeing significant demand (and shortages today) and there has been a flurry of investment activity in the space with larger players making strategic investments and acquisitions (e.g., JCI investing in Accelsius, Eaton acquiring Boyd). **In today's note, we focus specifically on cold plates.**

EXHIBIT 5: What exactly is liquid cooling for data centers?



Source: Bernstein Analysis

EXHIBIT 6: **Multiple specialized components needed for liquid cooling**

Equipment	CDUs	Cold plates	Manifolds	Refrigerant / coolant	Immersion tanks
Description	 Controls the volume and pressure of coolant flow through racks; also drives heat transfer of coolant to facility-side loop	 Metal plates that have refrigerant pumped into them; transfers heat through physical contact with GPUs	 Connects the CDU to the rack / cold plate system; effectively splits the flow of incoming fluid	 Liquid that drives the actual heat transfer; can be water + glycol mix or more specialized chemical refrigerants	 Physical equip. to submerge servers for immersion cooling; contact with coolant in tank drives heat transfer
Needed for...	Single-phase DTC	✓	✓	✓	Not needed
	Two-phase DTC	✓	✓	✓	Not needed
	Immersion cooling	✓ Design may evolve	Not needed	✓	✓
	Cold-plate etched DTC	✓	✓	✓	Not needed
	Silicon etched DTC	✓	Not needed (if channels etched into silicon)	✓	Not needed
Focus of this primer					

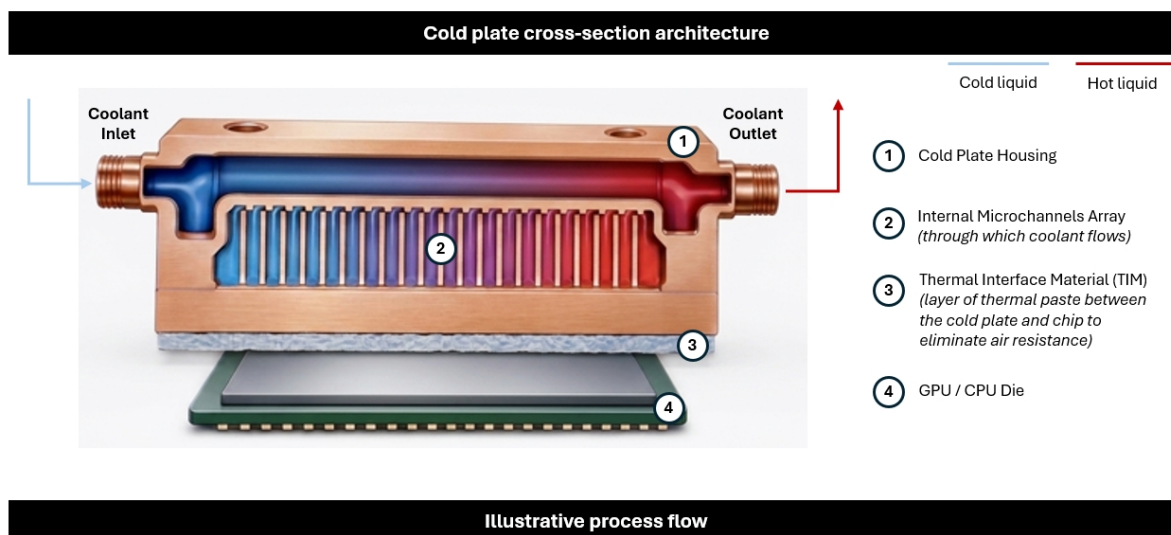
Source: Bernstein Analysis

WHAT IS A COLD PLATE?

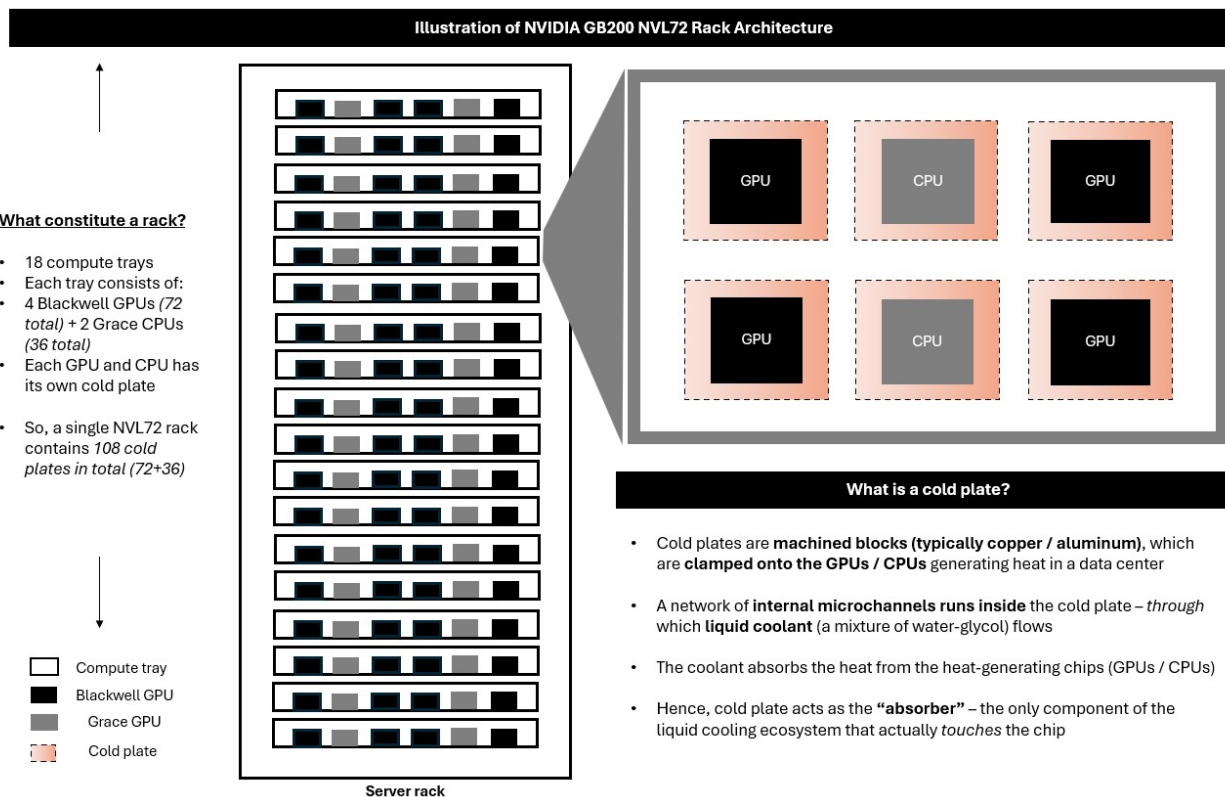
A cold plate is a machined metal block (copper / aluminum) which is clamped onto the GPU / CPU dies (the slot in which the actual chip architecture sits) on the silicon wafer. A network of minutely thin internal channels or fins (drilled through CNC / precision machining) runs inside the cold plate *through* which the liquid coolant (usually a mix of water-glycol pumped from the CDU in single-phase) flows. There is a layer of thermally conductive paste or pad between the cold plate and the GPU called the Thermal Interface Material (TIM). TIM serves as the thermal bridge between heat-generating GPUs and the cold plates - filling microscopic air gaps caused by surface roughness and ensuring consistent heat transfer from the GPU to the coolant (and thus the cold plate) through heat conduction. Even high performance liquid cold plates cannot achieve their full cooling potential if this interface is poorly managed, as hotspots can form due to uneven thermal contact. Post-heating, the warm coolant now exits back to the CDU, and thus back to the liquid cooling loop - and the cycle repeats.

In simple terms: the chip gets hot, the cold plate sits on top of it, the coolant flows through the cold plate's internal channels, carries the heat away as warm liquid, and comes back into the loop as cold fluid (post cooling from the loop). The cold plate acts as the “*absorber*” in this loop - it is the only component of the entire liquid cooling ecosystem that physically *touches* the chip.

The role of cold plates becomes clearer when viewed within the context of modern high-density AI server racks, such as NVIDIA GB200 NVL72 rack, which consists of 18 compute trays (each tray houses 4 Blackwell GPUs and 2 CPUs for a total of 72 Blackwell GPUs and 36 CPUs). Each GPU and CPU requires its own dedicated cold plate. Cold plates are thus mounted on each GPU / CPU die, resulting in a total of 108 cold plates per rack. Cold plates are inherently a high-volume component. Unlike CDUs (fewer in number and more complex), cold plates need to be deployed at scale, and must balance performance with ease of manufacturing.

EXHIBIT 7: **What is a cold plate?**

Source: Bernstein Analysis; Image generated using AI to display a directional cold plate cross section

EXHIBIT 8: **Server rack design and the role of cold plates**

Source: Bernstein Analysis

WHAT METRICS MATTER?

First, let's talk about what matters in a cold plate. There are dozens of technical specifications; while all matter in their own unique ways, we specifically call out five to keep an eye on:

1. **Thermal capacity:** Usually represented in W or kW, thermal capacity is the total rate at which a cold plate can remove heat from a chip (GPU / CPU) under operating conditions. It is influenced by coolant inlet temperature, plate design, thermal resistance, and channel geometry (discussed below). In hyperscaler deployments today, cold plates typically support ~2-5 kW per chip, with next-gen designs reaching 5-10 kW+ (CoolIT recently announced a 15 kW cold plate). As rapidly increasing AI workloads drive even higher power and rack densities, thermal capacity will be increasingly co-optimized with chip architecture going forward.
2. **Heat flux:** Measures the amount of heat removed from the chip per unit area in W/cm^2 , making it critical for understanding how well a cold plate can manage localized hotspots on the chip. This is particularly important for modern AI workloads, where heat generation is highly concentrated in small regions of the die where dense tensor cores compute the matrix multiplication for query resolution. Current hyperscaler targets are in the range of 100 - 130+ W/cm^2 , with leading-edge designs pushing towards 150+ W/cm^2 .
3. **Pressure drop:** This represents the amount of fluid resistance the coolant experiences as it flows through the cold plate, directly impacting the pumping power required to operate the system. As a result, lower pressure drop is preferred (it reduces the amount of pumping power) required to operate the system) as it reduces overall energy consumption and improves system efficiency. However, there is a tradeoff: achieving higher heat transfer (higher heat flux) increases resistance due to increased complexity in internal plate geometries (more thinly lined weaving channels). For current deployments, hyperscalers operate around 15-25 psi, with high performance designs reaching 30+ psi.
4. **Thermal resistance:** Quantifies the efficiency of the heat transfer from the chip to the coolant, expressed as a rise in temperature per unit of heat ($^{\circ}\text{C}/\text{W}$). Lower thermal resistance is desired, as this enables more efficient heat transfer, enabling the chips to run at lower temperatures for the same output. High-performance designs today run at $< 0.04^{\circ}\text{C}/\text{W}$. As chip power outputs continue to rise, solving for thermal resistance will become more critical, driving innovations in interface materials and surface-level engineering.
5. **Channel geometry:** Describes the internal structure of the cold plate, specifically the channel / fin density, size, shape, and arrangement of fluid channels, all of which together determine how effectively the coolant interacts with the hot surface of the GPU (a denser array = a higher surface area of contact between coolant and GPU = higher heat transfer). Complex geometries such as microchannels increase surface area and create greater turbulence (enables greater "scraping" of heat from the chip walls), thereby improving heat transfer.

HOW IS THE COLD PLATE ECOSYSTEM BEING SHAPED?

Cold plates are now a critical part of the liquid cooling ecosystem, as evidenced by recent announcements by NVIDIA and hyperscalers alike. For its next-gen Rubin architecture, NVIDIA has already mandated a "100% liquid cooling ecosystem with no fans anywhere", providing a secular tailwind for manufacturers in the space. By approving specific component vendors for different platforms through the Recommended Vendor List, NVIDIA is exerting soft influence on the evolution of the industry.

Unlike CDUs, there has been a relatively limited push from the industry to specify design configurations for cold plates so far. However, the OCP (Open Compute Project) has established baseline operational performance standards that an operational cold plate must meet (such as permissible pressure drop, leakage rates, operating temperature ranges, permissible amount of stress on the chip package, etc) which allows manufacturers to decide how exactly they want to achieve these performance metrics. We expect this to continue in the near-to-mid term while GPU TDPs continue to rise, enabling cold plate manufacturers to capture an innovation premium in their pricing (as they do with CDUs). However, longer-term, once TDPs stabilize and hyperscalers have a clearer idea of their compute portfolios we anticipate this innovation premium will erode as there is limited need for cold plate evolution. This does impact cold plates more than CDUs (which also see some commoditization risk) because they have limited to no service attach and this essentially puts the margin profile at risk.

EXHIBIT 9: What metrics matter in a cold plate?

Metric	1 Thermal capacity	2 Heat flux	3 Pressure drop	4 Thermal resistance	5 Channel geometry
Question answered	How much total heat can the plate remove from the chip?	How much heat can the plate remove from the chip per unit area?	What is the fluid resistance coolant encounters while inside the plate?	How efficiently is heat transferred to the coolant?	How much is the coolant physically interacting with the surface?
Description	Rating of the cold plate describing the total amount of heat dissipated per GPU / CPU; driven by chip TDP + rack density Higher is better; depends on chip architecture, plate mounting conditions	Rating of the cold plate describing the maximum amount of heat dissipated per GPU / CPU; driven by chip TDP Higher is better; but geometry needs to be scalable with low failure rates	Amount of pressure loss in the coolant across the plate array Lower is better; high resistance implies more pumping power (from the CDU) and also increases clogging risk	Measures the efficiency of the heat transfer path Lower is better as chip power increases, the temperature difference b/w chip and cooling water shrinks; hence thermal resistance needs to be low	Fin density and channel shape determine surface area and flow turbulence Higher is better; but complex geometries increase pressure drop, hence optimizing design is a constant tradeoff
What does "good" look like?	Measured in W / kW ; while current-gen supports 2-5 kW per chip, next-gen plates push 5-10kW+	Measured in W / cm² ; hyperscalers now look for 150+ W/cm²	Best in class 30+ psi for high flux designs ; around 15-25 psi for widely available designs	Measured in °C / W ; best-in-class < 0.04°C / W	Microchannels / fins optimized for turbulent flow; to "scrape" heat of the chip walls

Source: Bernstein Analysis and Estimates (for power ratings)

EXHIBIT 10: Different geometry types inform customization in cold plates

Compressor type	Microchannels	Internal Strip Fins	Machined Serpentine	Pressed Tube
Operating principle	 Straight channels of up to 1mm width; extremely dense array	 Individual fins inside the plate; create some turbulence	 Continuous loops machined through plate body	 Copper / stainless tubes pressed / brazed into the plate
Heat flux	Potentially more than 200 W/cm ² (highest)	~120 W/cm ²	~60 W/cm ²	~20-40 W/cm ²
Market share	~40- 50%	~20 -25%	~15 - 20%	< 5 -10%
Manufacturing complexity	High	Moderate	Moderate-low	Low
Commentary	Used in high-power GPUs for hyperscalers (highest surface area for heat absorption) Key focus of innovation	Mainly used in mainstream CPU deployments / early-stage GPU applications	Used in medium-power CPUs, legacy and retrofit deployments	Entry-level cooling architecture used in niche, non-AI applications

← Unlike Coolant Distribution Units (CDUs), hyperscalers do not specify cold plate geometries; instead defining only thermal performance requirements that a cold plate must meet →

Source: Bernstein analysis and estimates (heat flux, market share)

EXHIBIT 11: Cold plate designs not mandated; specifications and outcomes are

Open Compute Project (OCP)	NVIDIA Recommended Vendor List
1 OCP is a non-profit (Meta, Google, Microsoft, Oracle as members) that has published open-source specifications for data center hardware 2 As part of OCP, Google has published Project Deschutes (2025, technical specifications for liquid-to-liquid cooling CDUs) & Project Brazos (2026, liquid-to-air CDU specifications for retrofit deployments) 3 Unlike CDUs, OCP has <i>not</i> specified design parameters (like a channel array pattern) for cold plates. Instead, they define a set of operational performance standards that a cold plate must meet: - Allowable pressure drop - Operating temperature range - Allowable leakage rates - Allowable stress on chip package 4 Hence, while OCP states the broad playbook (in terms of base parameters) for cold plate design, manufacturers have a free hand on designing internal geometry of the plate	1 NVIDIA has stated a “100% liquid cooling” ecosystem for Rubin architecture (..”every networking component, cooled entirely by liquid in a closed loop with no fans anywhere in the system”..) 2 NVIDIA wants multiple cold-plate suppliers: to derisk supply chain and scale volumes 3 Announced “Recommended Vendor List” – list of approved vendors designing components specifically for NVL72 rack architecture (e.g., Boyd, CoolIT) 4 Strategic partnership with Vertiv - NVIDIA and 5 partners won \$5 million US Department of Energy COOLERCHIPS grant in 2023 to build advanced liquid-cooling systems - Partners include Vertiv, Boyd, Binghamton University and Villanova University - The team intends to develop an integrated system deploying two-phase DTC cooling (using cold plates) and immersion cooling <i>in tandem</i>

Source: Bernstein Analysis, Company Reports

EXHIBIT 12: GPU heat generation is non-uniform; cold plates engineered to target hotspots are the next innovation frontier

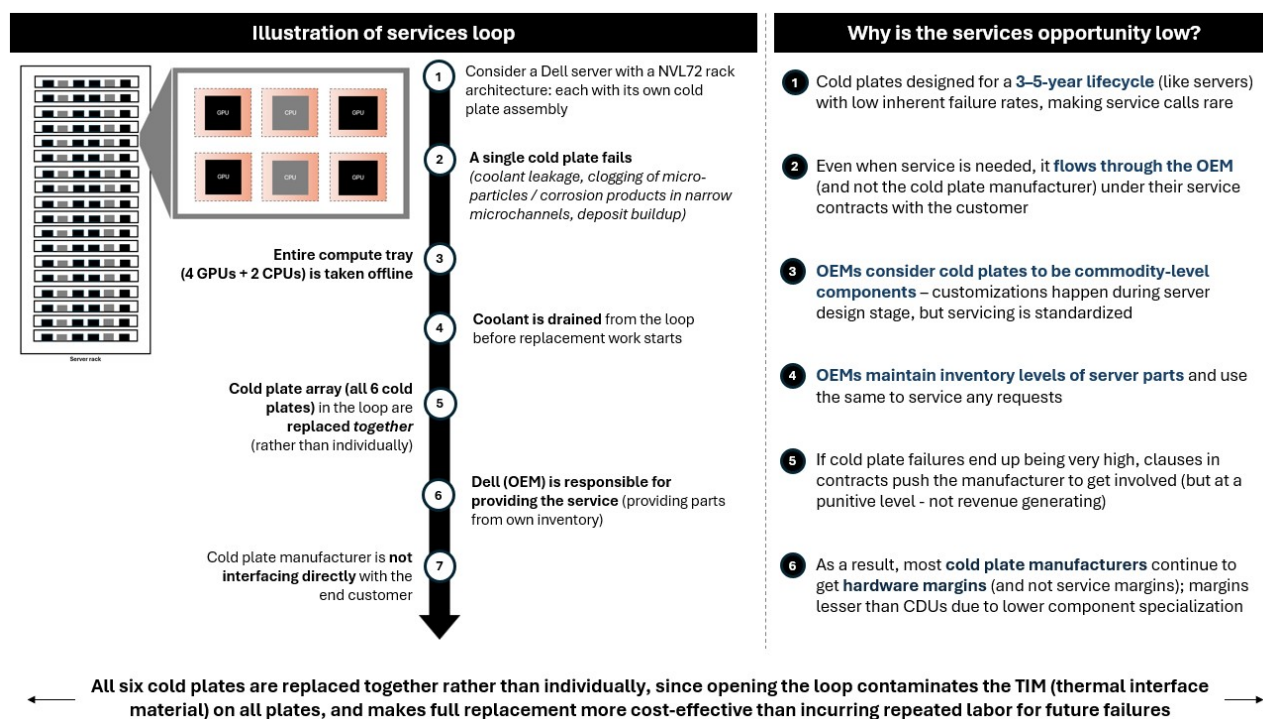
	Description	Illustration
Non-uniform heat generation in GPU	- A GPU is carved into different functional regions – with dense compute blocks (tensor cores that do the matrix math for AI) drawing the highest power - GPU power density (W/cm²) is highly non-uniform: compute-intensive regions create localized hotspots (typically near the center) – center blocks can be 2-3x hotter than average Increasing AI workloads amplify hotspot intensity; pushing traditional uniform cooling approaches to theoretical limits	GPU Heat Map
Custom-made cold plate designing	- Cold plates are engineered to match GPU heat map – with high cooling capacity concentration over hotspot centers - Non-uniform coolant flow designs (curved flow routing) direct maximum coolant flow (and hence high heat transfer) to the hottest areas Hotspot-optimized custom designs thus enable higher rack densities; making them a key enabler of next-gen data center performance	Cold plate cross-section

Source: Bernstein Analysis

WHY DON'T COLD PLATES HAVE A GREAT SERVICE ATTACH?

Cold plates are largely designed to last for the lifecycle of the GPUs they are cooling. However, there is naturally unexpected failure that takes place in the racks. When this happens, individual cold plates do not get “serviced” and re-attached, they are simply replaced. This is generally done by the server OEM (companies like Dell) which keep an inventory of cold plates and charge for the service of making a swap. This is less intensive work vs. repair or maintenance where pieces of equipment are checked and tested. In addition, for the cold plate manufacturer themselves, they only receive hardware margins (which look set to compress as cold plates themselves have commoditization risk). There are some situations in which the cold plate OEM themselves become responsible for service, but that is more contractual when failure rates exceed acceptable levels and they have a contractual obligation to step in and manage the swaps; terms are not favorable and no OEM would actually want to be in this situation. This is fundamentally different from a CDU where there are multiple components in the unit (pumps, heat exchangers, etc.) that could fail and need maintenance and repair. CDUs also have longer useful lives which allows greater monetization of the install base.

EXHIBIT 13: Service attach structurally limited for cold plates



Source: Bernstein Analysis, Expert Interviews

COMPARISON OF KEY PLAYERS IN THE MARKET

Naturally, as liquid cooling has accelerated, many players have entered the market. The cold plate manufacturing market is characterized by a mix of established industrial players and emerging startups. Large incumbents (such as Boyd, CoolIT, CoolerMaster) bring scale, manufacturing expertise and established hyperscaler relationships, while funded startups focus upon specialized innovations like cold plates for two-phase DTC cooling ecosystem.

Given the opportunity, there has been a spree of M&A activity in the space, with leading manufacturers of electrical and HVAC equipment (JCI, Carrier, Eaton, Schneider) wanting a slice of the AI infrastructure pie. As volumes scale and hyperscalers increasingly consolidate their supplier bases, we expect competitive dynamics to shift towards a balance between manufacturing scalability and technical differentiation.

EXHIBIT 14: **Combination of mature players and startups have cold plate exposure**

Company							
Year founded	1928	2001	2014	1992	1988	2022	2016
Country							
Funding / acquisition price	\$9.5 billion (acquired, ~5.6x 2026 revenue)	\$4.75 billion (acquired)	NA	Self-funded	\$850 million (acquired, mid-single digit 2025 revenue Mx)	\$89 million (2 rounds)	\$111 million (3 rounds)
Key investors / acquirers	Eaton	Ecolab	Vertiv	Private	Schneider Electric	Johnson Controls, Legrand	Carrier, Mitsubishi, Samsung
Cold plates % of revenue	~40-50%	~40%	Mix b/w cold plates & heat exchangers	Small, but growing	Not available	Split b/w cold plates, CDUs, manifolds	Split b/w cold plates & CDUs
Commentary	Acquired by Eaton in Mar 2026 from GSAM Largest player globally having delivered 5 million+ cold plates to hyperscalers as of Sep '25	Acquired by Ecolab in Mar 2026 from KKR (15x return in 3y) Ecolab rationale to become a 'one-stop shop cooling solutions provider' to hyperscalers	Acquired by Vertiv in Apr 2026 Focus on high chip-density thermal problems, filling the gap between chip and facility-level cooling	33+ years of thermal manufacturing expertise based out of Taiwan Primarily sell to server OEMs instead of directly to hyperscalers	Schneider acquired 75% stake for \$850M Patented 'wide groove' tech with demonstrated capacity of up to 1000 W per GPU (distribute fluid top-down simultaneously across the chip)	Focused exclusively on two-phase cold plates with capacity of up to 4500 W per GPU Exclusively targeting next-gen NVIDIA Kyber rack architecture (vertically oriented server racks)	Waterless, two-phased DTC cooling system with capacity of up to 600 W / GPU 'No pump' approach (unlike Accelsius) where phase change drives circulation by gravity differential

Backed startups focused exclusively on two-phased DTC

Source: Bernstein Analysis, Company Reports

SIZING THE MARKET

We have used a top-down estimate to size the overall market for cold plates using GW added as a starting point. While we do have a base / bull / bear case for cold plates (see Exhibit 10), these are based on certain assumptions (laid out below). While the paucity of available data and pace of change are a challenge, we have used our judgment to develop a robust point of view.

First, we look at new GW capacity being added each year. This creates a thermal baseline in terms of power consumed. Most of that power is released from chips. with an additional factor (1.2x multiplier in our model) to account for replacement / repair-based demand. Next, we estimate the % share of data center capacity that is liquid cooled. The number evolves from a 20-30% range right now to becoming the majority share in our cases by 2030 (evidenced by hyperscaler / NVIDIA commentary). Next, we assume the total number of GPUs per GW (as a blended capacity added amongst Blackwell / Hopper GPUs in 2026, and early Rubin architecture for 2030). The number of cold plates per GPU is estimated to be 1 (both today and in the future). Our research suggests an ASP per unit of \$150-300 currently (basis design complexity for low - high TDP GPUs), which translates into a \$/kW cooling factor of 0.30 (500-1000 kW TDP per GPU assumed). We model the cooling factor of 0.20 \$/kW and 0.25\$/kW in the base and bull cases respectively to bake in potential commoditization of cold plates going forward (thus arriving at the ASP per cold plate by multiplying with estimated TDP per GPU from 2000-3000kW in 2030). For the bear case, we assume cold plate prices stay the same. This implies a market size of \$2 - 3B in 2026 and \$6 - 7B in 2030 (base case), implying a 4Y CAGR of ~20-30%.

Naturally, this model is sensitive to any assumptions you may have (especially GW added which can vary substantially between sources, AND ASP per cold plate). The benefit of our approach is that one can relatively easily modify these assumptions and see what the market would be accordingly. **Please reach out to the authors or your Bernstein sales counterpart for the bottom-up market size buildout.**

EXHIBIT 15: Top-down cold plate market sizing

Year	2026E	2030E (Bear)	2030E (Base)	2030E (Bull)
GW added in year ¹	~20 GW	~40 GW	~40 GW	~40 GW
	×	×	×	×
% share of liq. cooling	~30 - 40 %	~50 - 60 %	~60 - 70 %	~70 - 80 %
	×	×	×	×
# of GPUs / GW ²	~1 - 2 M	~0.3 - 0.5 M	~0.3 - 0.5 M	~0.3 - 0.5 M
	×	×	×	×
# of cold plates / GPU	1	1	1	1
	×	×	×	×
Multiplier ³	1.2	1.2	1.2	1.2
	×	×	×	×
ASP per unit ⁴	~\$150 - 300	~\$150 - 300	~\$400 - 600	~\$500 - 750
	▼	▼	▼	▼
Cold Plates Market Size	\$2 - 3B	\$2 - 3B (flat CAGR)	\$6 - 7B (~20-30% CAGR)	\$8 - 10B (~35-40% CAGR)

1. Sourced from McKinsey 2. Assuming a range of 500-1000kW per GPU (split among Blackwell/Hopper) for 2026, range of 2000-3000kW per GPU (basis Rubin estimated) for 2030 3. To account for replacement / repair-based demand to materialize by end of decade 4. Estimated basis 0.30 \$/kW cooled (2026), flat prices (2030, bear case), 0.20\$/kW (2030, base case), 0.25\$/kW (2030, bull case); we have built a conservative estimate of a decline in \$/kW to bake in potential commoditization of cold plates

Source: Bernstein Analysis and Estimates (all inputs)

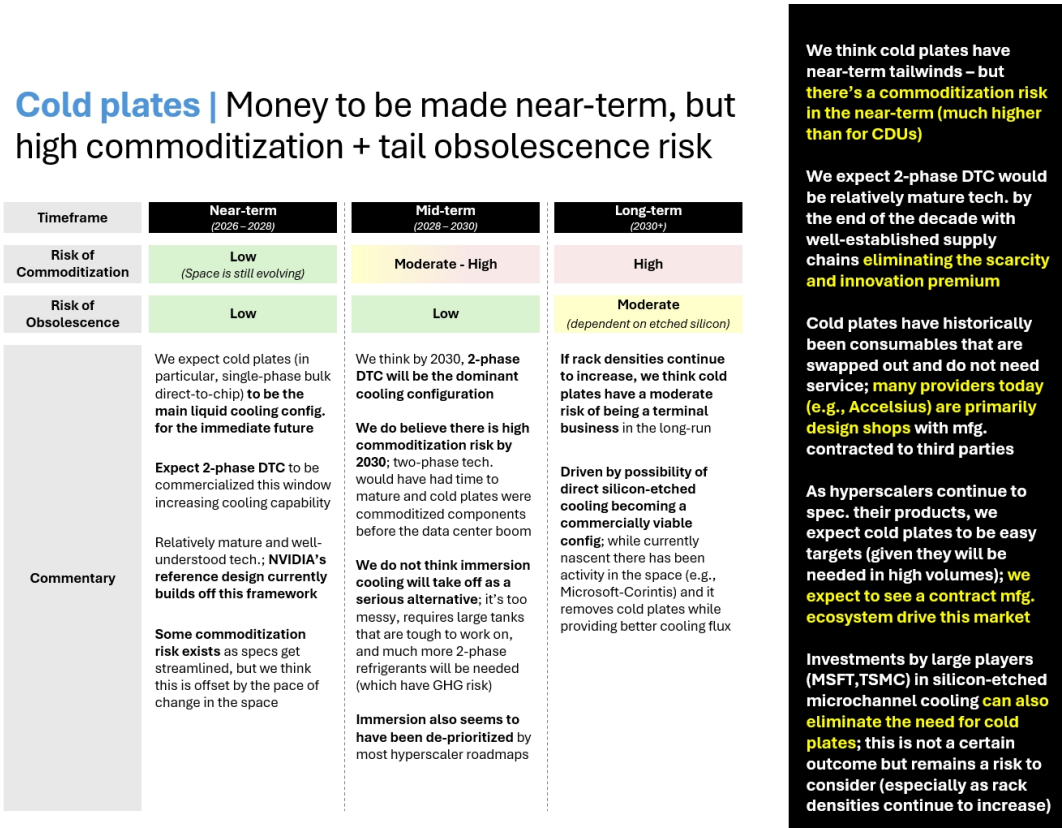
LOOKING AHEAD

Overall, we are bullish on the outlook for cold plates in the near-term, but advise a cautious view for the medium to long-term. Driven by the rapid scaling of AI workloads, the corresponding increase in GPU thermal design power (TDP) and rack densities - we expect direct-to-chip (DTC) liquid cooling to become the de-facto mode of data center cooling (further supported by relative technological maturity, scalability, adoption, and compatibility with existing infrastructure). Cold plates are an integral component of the liquid cooling ecosystem, and are thus expected to grow in terms of usage in the near-term.

Unlike more complex system components, cold plates offer limited opportunity for recurring services revenue. They are typically built for a lifecycle aligned closely with that of the server itself (3-5 years), and failure rates are relatively low. In the of chance that a failure does occur (due to coolant leakage, clogging or deposit buildup in the internal channel array), servicing requires taking the entire compute tray offline (see Exhibit 12), and replacing all the constituent cold plates. Servicing is (usually) carried out by the server OEM itself (rather than the cold plate manufacturer) as a part of their annual service contract with the customer. This dynamic reinforces the characterization of cold plates as a primarily hardware-driven revenue stream, with limited post-sale monetization for the manufacturer.

The medium to long-term outlook for cold plates also has some structural risks. A primary one is a potential commoditization of cold plate designs as single-phase and two-phase DTC systems become more mature and widely adopted. We've heard from cold plate manufacturers themselves that the value of cold plates sits in the design, not in manufacturing (with some players using contract manufacturers themselves). Additionally, emerging technologies such as silicon-etched cooling and direct-to-die cooling could eliminate the need for cold plates entirely by directly integrating cooling into the chip package (which is not the case with CDUs). While still in nascent stages, such developments represent a credible long-term disruption vector that could define the course of the space.

EXHIBIT 16: Cold plates attractive in near-term, but see real commoditization and obsolescence risk



Source: Bernstein Analysis

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